Sequence representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why can two different image regions have the same pooled maximum?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Encode an ordered sequence with an explicit vocabulary. Try the recall before opening notes.

Your learning target

I can encode an ordered **sequence** with an explicit vocabulary.

I can use **padding** and a **mask** without inventing real events.

I can identify information lost by counts unknown symbols and truncation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Encode an ordered **sequence** with an explicit vocabulary

Use **padding** and a **mask** without inventing real events

Identify information lost by counts unknown symbols and truncation

Retrieve what you already know

> Why can two different image regions have the same pooled maximum?

> **My starting explanation**

This week's end goal

By the end, I can explain **sequence** representations using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Sequence representations case. Separate observations, given facts and assumptions before choosing an explanation.

Use fictional robot event logs whose symbols are GO, TURN and STOP. The task asks whether the final real event is STOP. This is a representation exercise, not a claim that an event log proves a robot is safe.

GO then STOP and STOP then GO contain the same symbols but have different final events.

Log	Counts in GO TURN STOP order	Final event	Task answer
GO STOP	[1, 0, 1]	STOP	Yes
STOP GO	[1, 0, 1]	GO	No
GO TURN STOP	[1, 1, 1]	STOP	Yes

A count vector discards order, so no deterministic rule using only the first two count vectors can distinguish their answers. A **sequence** of IDs preserves order if we retain each position.

Declare the vocabulary: PAD = 0, UNK = 1, GO = 2, TURN = 3 and STOP = 4. PAD is reserved for added storage positions; UNK represents a real but unrecognised symbol.

IDs are names, not measured amounts. STOP having ID 4 does not make it twice as much as GO with ID 2. Changing the vocabulary requires changing every consumer consistently.

Do not interpret numerical distance between IDs as semantic similarity.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Sequence representations. Connect each labelled input, deciding step and result.

Store each log in four positions, **padding** on the right. A **mask** value 1 marks a real position and 0 marks **padding**. The encoder rejects empty logs, logs longer than four, and literal reserved markers.

Other unrecognised symbols become UNK and still receive **mask** 1.

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

The code assumes a list of symbol strings and uses exact spelling. It does not fit a vocabulary from data.

If a future project fits vocabulary choices or frequency cutoffs, derive those choices from **training** data and apply the saved encoding consistently to validation and final cases.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Sequence representations. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: A missing **sequence** length causes every numeric0 to be treated as **padding** in the sensor example.

Predicted result

Prediction: The first real sensor reading is incorrectly discarded.

Why the result follows

Reason: Its numeric value is 0, but its position is real according to the declared length.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Sequence representations to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Sequence	An ordered list of elements in which positions can matter
Token ID	A numerical identifier assigned to a symbol by a vocabulary
Padding	Placeholder positions added to make stored sequences the same length
Mask	A list marking which stored positions are real inputs and which are placeholders

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Sequence representations, predict a step, then compare with its explanation.

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sequence representations

Distinguish **padding** positions from meaningful zero sensor readings.

The facts we will use

Supplied facts

Toy tokensPAD0, UNK1, LEFT2, RIGHT3; stored length4. **Sequence**LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/25

Sequence representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Encode real tokens

LEFT and RIGHT occupy positions0 and 1.

2. Mark padding

Positions2 and 3 are storage padding; mask marks them 0.

3. Compare sensor example

Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

CHANGE ONE THING

A missing sequence length causes every numeric0 to be treated as padding in the sensor example.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode real tokens	LEFT and RIGHT occupy positions0 and 1.
2. Mark padding	Positions2 and 3 are storage padding; mask marks them 0.
3. Compare sensor example	Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

What this example does not establish

Keep the stated limit

Token **padding** rules need not apply to sensor values. Document **sequence** lengths,

vocabulary and unknown-token handling.

Keep the stated limit

Read the labelled picture: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A missing **sequence** length causes every numeric0 to be treated as **padding** in the sensor example. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sequence representations claim and explain a repair using the supplied evidence.

Check the claim against the facts

A **mask** must affect a calculation to protect it from **padding**. Suppose real sensor values are [2, 4], stored as [2, 4, 0, 0] with **mask** [1, 1, 0, 0]. The **mask**ed mean is (2 + 4) / 2 = 3. Averaging all four slots gives 1.5.

Unlike arbitrary **token ID**s, these sensor values are measurements for which a mean is meaningful.

Check the claim against the facts

Measured zero must remain real: values [0, 4, 0, 0] with **mask** [1, 1, 0, 0] have mean 2. Removing every zero would incorrectly give 4. The **mask** distinguishes a measured zero from a storage placeholder.

An all-zero **mask** has no real values; reject it or follow an explicitly defined empty-**input** policy instead of dividing by zero.

Storing ordered IDs makes order available but does not guarantee a later model uses it. A computation that sums token counts or averages position-independent vectors can lose order again.

Trace what the whole procedure reads, not just what the **input** format could preserve.

Locate the earliest unsupported step

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Sequence representations with support, then explain the deciding step in your own words.

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

Explain your prediction to a partner before comparing results.

Practise with less help: Expose an order collision

Use workbook W2 for the connected example "**Sequence** representations". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Sequence** representations. Proposed explanation: "Larger **token ID**s mean more of a concept". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
>
>
>
>
>

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Sequence representations. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sequence representations. Explain why each step is justified.

Explain → apply → challenge

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **sequence** representations **evidence** sheet to Image-pattern Network Investigation. Show how you can encode an ordered **sequence** with an explicit vocabulary.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

WAIT STOP and PAUSE STOP both become [1, 4, 0, 0] with the same **mask**. UNK retains a real position but loses the identity of the unrecognised symbol.

The final-event task can still answer Yes for both because the last real ID is STOP. A final real UNK would require an unknown result or another documented policy when the intended event cannot be determined.

> **A log GO TURN GO TURN STOP is cut to its first four events before encoding. What happens to the final-event answer? Explain why the lesson encoder rejects the original log instead of silently truncating it.**

The full log ends in STOP, while the retained prefix ends in TURN. Truncation changes the **evidence** needed for this task. Save the original length and any truncation policy in a real experiment, and test the con**sequence**s.

Next week introduces embeddings while preserving the distinction between identifiers and learned vectors.

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sequence representations case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a fresh schema independently

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Sequence representations, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Distinguish **padding** positions from meaningful zero sensor readings.

By the end, I can	How I will show it
Encode an ordered sequence with an explicit vocabulary	Explain the deciding step, show a trace or test, and state one limit.
Use padding and a mask without inventing real events	Explain the deciding step, show a trace or test, and state one limit.
Identify information lost by counts unknown symbols and truncation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sequence representations without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Encode an ordered **sequence** with an explicit vocabulary

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish **padding** positions from meaningful zero sensor readings.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

1. Encode real tokens

2. Compare sensor example

3. Mark **padding**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare sensor example Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A missing **sequence** length causes every numeric0 to be treated as **padding** in the sensor example. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The first real sensor reading is incorrectly discarded.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Repair a missing building block

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

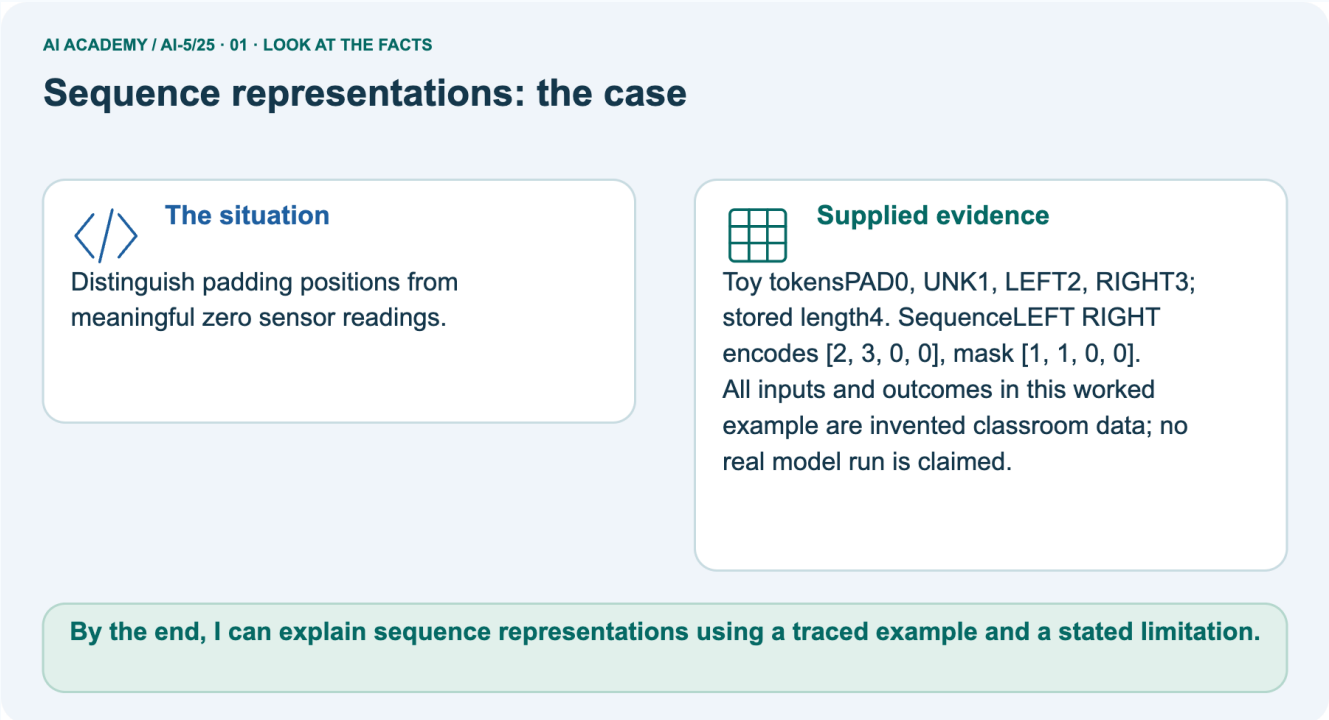
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sequence representations graphic using its labels, deciding evidence and relevant limitation.

01. Sequence representations: the case



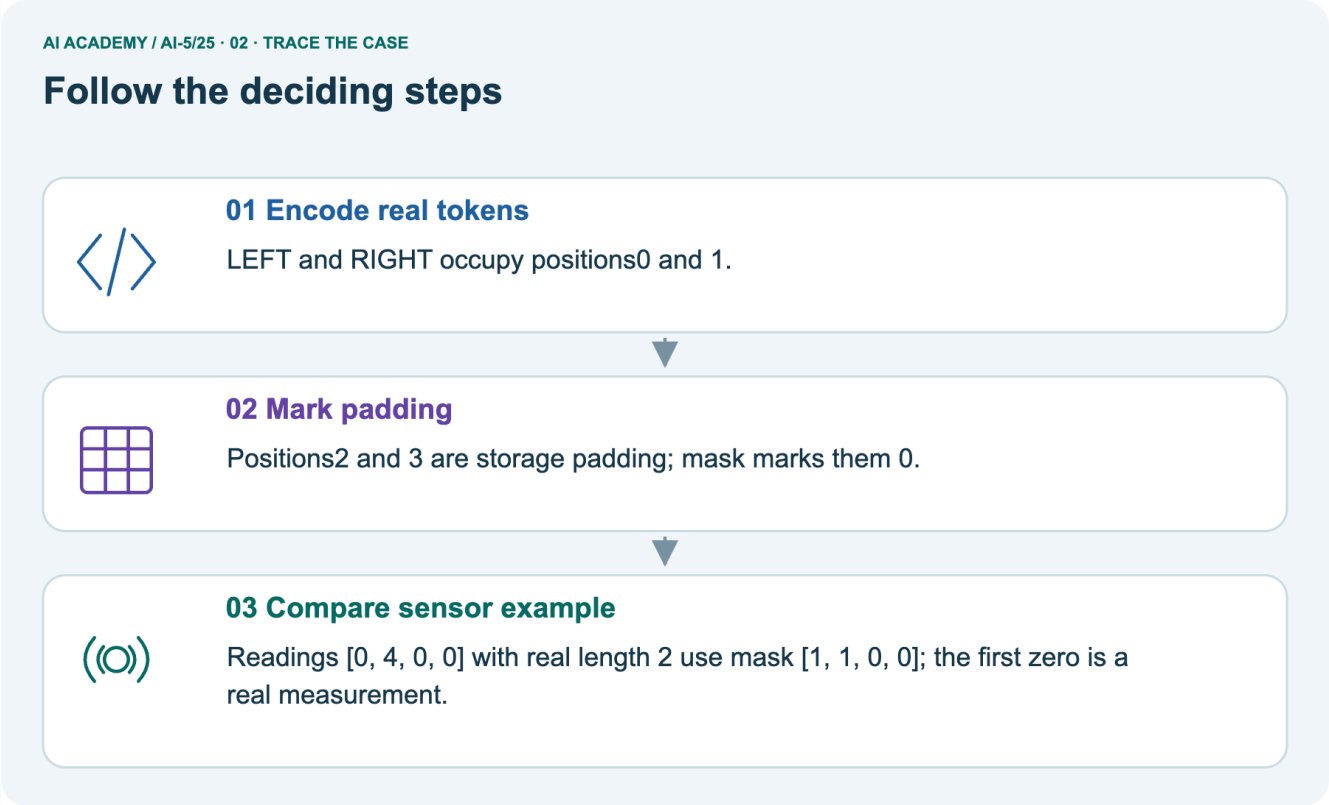
Follow the labelled picture with the matching explanation. Open the vector diagram

Distinguish **padding** positions from meaningful zero sensor readings. Supplied facts: Toy tokensPAD0, UNK1, LEFT2, RIGHT3; stored length4. **Sequence**LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Encode real tokens: LEFT and RIGHT occupy positions0 and 1. → Mark **padding**: Positions2 and 3 are storage **padding**; **mask** marks them 0. → Compare sensor example: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Point to each numbered stage and explain which recorded **input** it uses.

03. A padded log still has two real events

AI ACADEMY / AI-5/25 · 03 · INSPECT THE LESSON MODEL

A padded log still has two real events

Stored position	Token ID	Mask
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

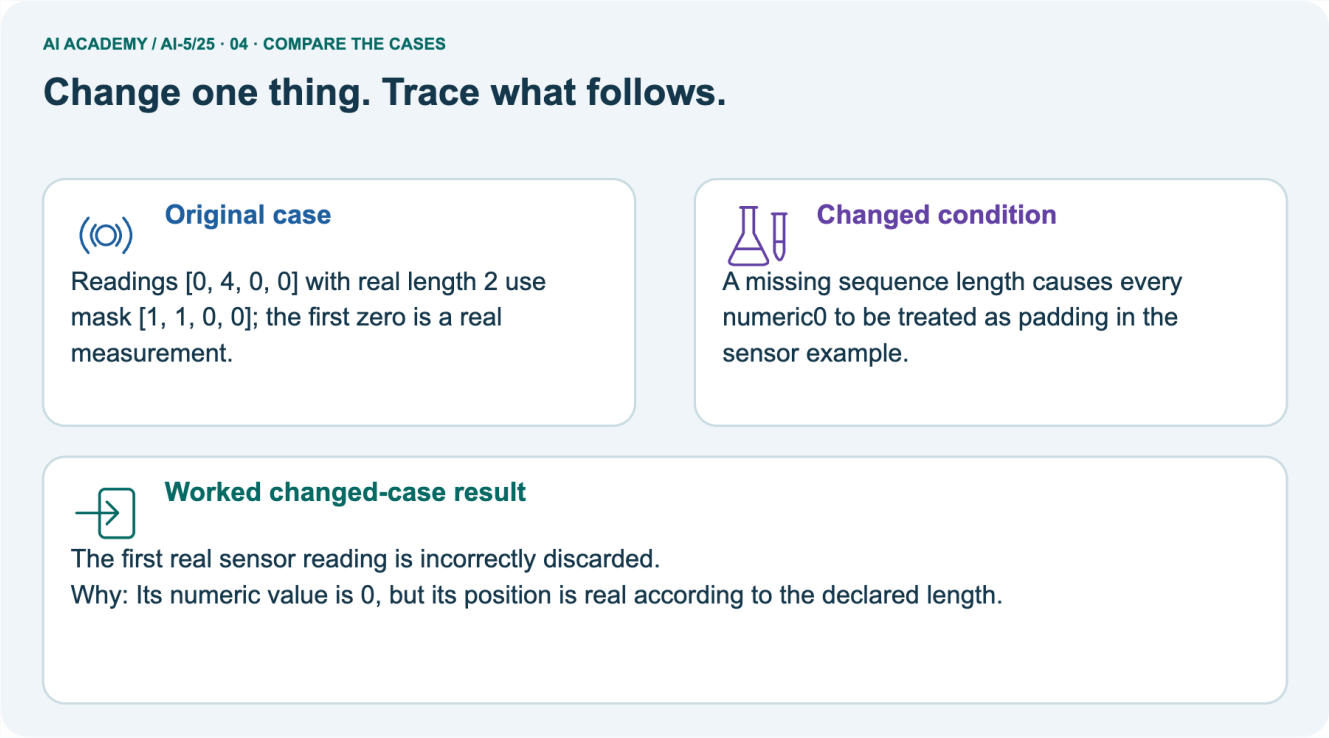
The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

A padded log still has two real events. 0 | 2 meaning GO | 1 real. 1 | 4 meaning STOP | 1 real. 2 | 0 meaning PAD | 0 placeholder. 3 | 0 meaning PAD | 0 placeholder

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

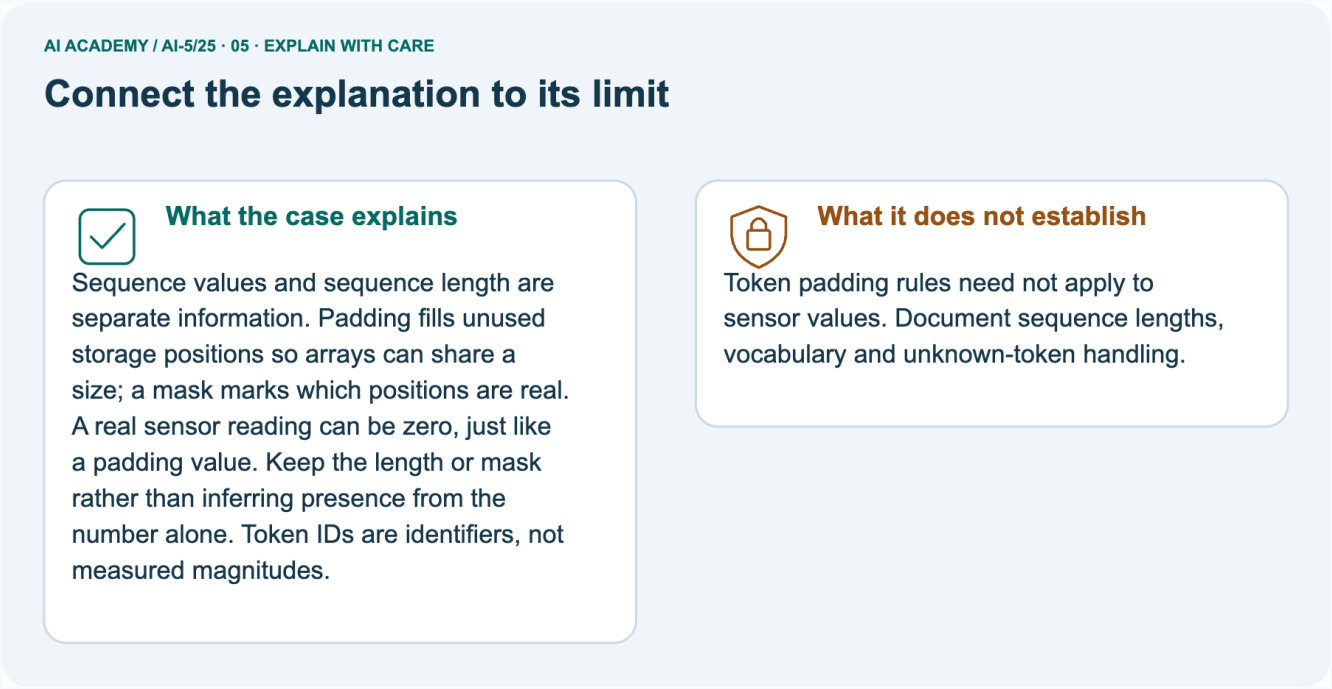
Original case: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Changed condition: A missing **sequence** length causes every numeric0 to be treated as **padding** in the sensor example. Expected result: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

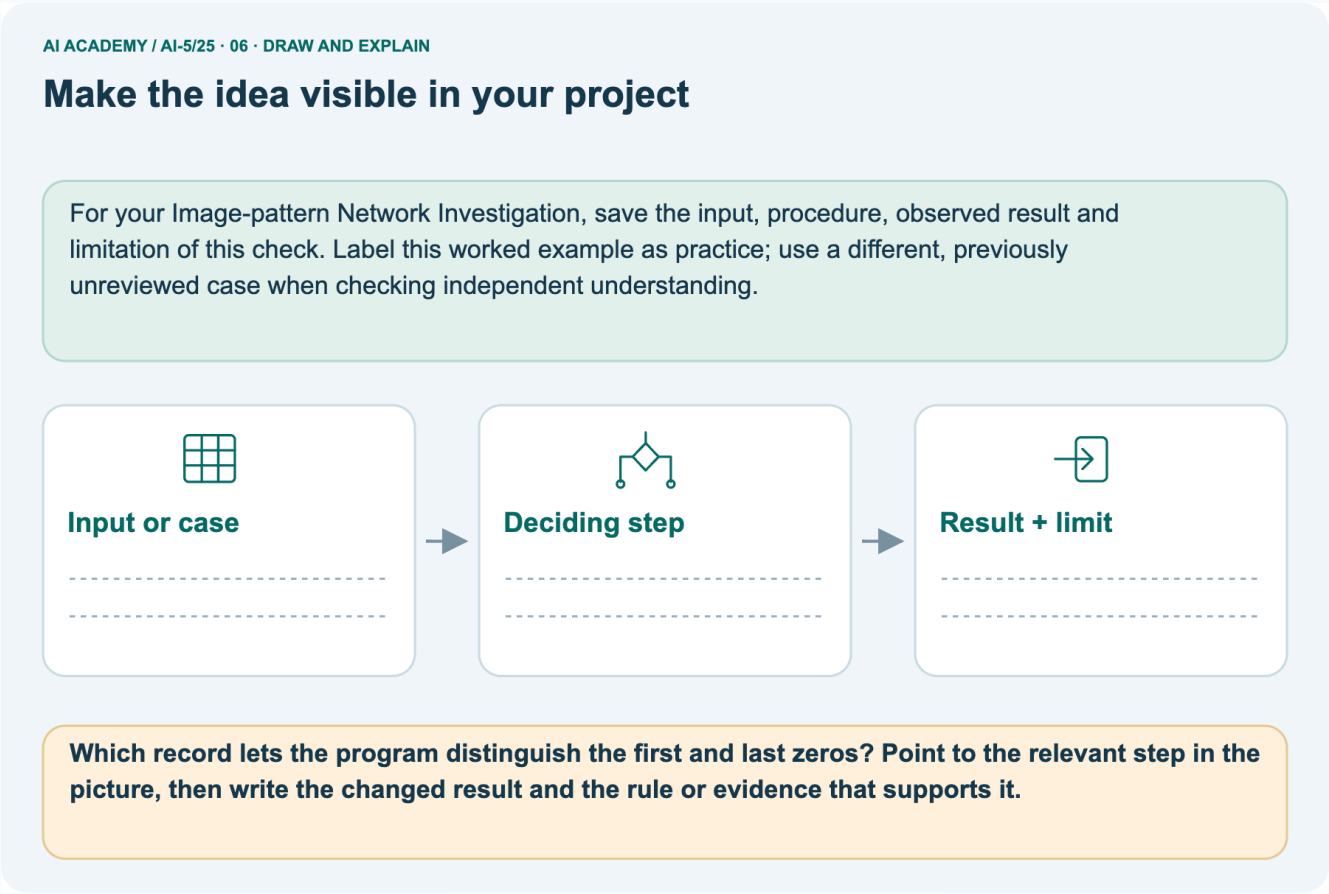
Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Limit: Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.